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AI-Augmented Memory and Consumer Sustainability: Understanding Dual Consumption Pathways in Digital Retail

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ABSTRACT

In digital retail, algorithmic personalisation shapes consumer outcomes, yet the cognitive mechanisms involved in this process remain theoretically underspecified. This paper introduces AI-augmented memory (AI-AM), a mechanism that algorithms use to record, retrieve and resurface prior behavioural traces during consumer decision-making. The study draws on 30 semi-structured interviews with Singapore-based e-commerce consumers and a complementary practitioner-reflexive account by the researcher, based on over 10 years of multi-platform retail experience. It examines how AI-AM both enables and undermines SDG12 (responsible consumption and production). The thematic analysis identifies a dual pathway: a sustainable pathway, where AI-AM reduces trial-and-error purchasing and supports deliberative restraint and an unsustainable pathway, where the same mechanism amplifies impulse purchase loops and erodes autonomous restraint. Four contextual conditions were associated with pathway activation: a pragmatic need-based norm, platform-mediated social proof, age-associated variation in pathway responses and reliance on AI recommendations. The study extends the Engel–Kollat–Blackwell (EKB) model by theorising AI-AM as a recursive dual-pathway sustainability gate and develops a sustainability-aligned design typology for platform designers and policymakers.

1 | Introduction

Consider two consumers, both on Shopee at 10 p.m. The first person receives a push notification. A product she added to her cart 3 weeks ago has dropped in price. She pauses to consider if she still needs it and then buys deliberately. The second person scrolls through a personalised feed and, within 7 min, buys three items he had not intended to purchase. An AI-curated flash-sale banner, a 'customers also bought' carousel and an AI-generated review summary all influence these purchases. Although both experiences use the same algorithmic architecture, their impacts on sustainability differ significantly.

This paper argues that this difference is deliberate. The AI-augmented memory (AI-AM) mechanism that algorithms use to resurface prior behavioural traces can trigger either a sustainable or an unsustainable consumption pathway. This depends on how it is used and how the consumer responds. This study focuses on understanding which pathway is activated and why.

The intersection of artificial intelligence and sustainable consumption has become an urgent research frontier and a strategic priority. Global AI-driven data centre electricity consumption is rapidly growing, with the IEA estimating that it could double by 2030 (International Energy

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Agency 2025). Furthermore, the European Green Deal (European Commission 2019) and the EU AI Act (European Parliament and Council of the European Union 2024) have intensified policy attention on transparency, accountability and sustainability in AI-enabled markets. Against this backdrop, understanding how AI systems shape sustainable consumer behaviour is no longer a niche academic interest. It has become a governance imperative.

So far, existing research remains limited. The literature on AI in marketing (Grewal et al. 2025; Mariani et al. 2022) and the literature on sustainable consumption (White et al. 2019; Bergianti 2025) are largely independent of each other. Where they intersect, many studies have examined how AI influences shopping at the cognitive level, exploring consumer trust, perceived control, attitudes and experiential responses to algorithmic systems (Puntoni et al. 2021; Hermann et al. 2024). However, this research has not specifically investigated *memory*. Consumer memory has been established as essential for decision-making (Alba et al. 1991; Blackwell et al. 2001), but it is viewed as an internal, passive repository rather than as a resource that platforms actively record, select and resurface. Thus, this paper specifically addresses the following research gap: how platform-initiated reactivation of consumer behavioural traces, the AI-AM mechanism introduced above, shapes the conditions under which sustainable or unsustainable consumption decisions are made.

The study shows that AI-AM functions as a memory-based sustainability mechanism influencing purchasing decisions over time, rather than a single purchase. The sustainability of outcomes depends on three factors examined in this paper: the specific AI-AM mechanism activated, the consumer's autonomy stance when influenced by AI and the cultural and demographic context in which AI-AM operates.

The study makes three theoretical contributions. First, it introduces AI-AM as a dual-pathway sustainability mechanism that goes beyond cognitive efficiency to include its effects on sustainability. Second, it reconceptualises the memory stage of the Engel–Kollat–Blackwell (EKB) model as more than a neutral cognitive resource. Instead, it is viewed as a contested sustainability node influenced by algorithms with varied consequences for sustainability. Third, it identifies four context-specific patterns linked to pathway activation in Singapore: pragmatic need-based norms, platform-mediated social proof, age-associated pathway variation and reliance on AI recommendations. These findings should be tested in future cross-cultural research, with direct implications for platform design and regulatory policy in Asia-Pacific digital markets.

The paper proposes a sustainability-aligned AI-AM design typology to assist platform designers and regulators in configuring AI memory systems that support SDG12 goals.

2 | Literature Review and Theoretical Development

Three areas of research define the paper's theoretical contribution: (1) AI's impact on consumer memory and decision-making, (2) AI's role in both advancing and undermining sustainable consumption and (3) the conditions that affect AI's influence on consumer behaviour.

2.1 | AI-Augmented Memory and Consumer Decision-Making

2.1.1 | Memory in Consumer Behaviour: The Classical Foundation

Consumer memory has long been seen as a key part of decision-making. In the EKB model (Engel et al. 1968; Blackwell et al. 2001), memory serves as the repository from which information is retrieved during the information search stage. Lynch and Srull (1982) showed that memory-based choice, where consumers retrieve and evaluate alternatives from memory, differs from stimulus-based choice where external cues drive retrieval. This distinction was developed in a pre-digital time when memory was seen as an internal, individually managed cognitive resource.

The digital transformation of retail has disrupted this assumption. Consumers on AI-mediated platforms do not encounter neutral information. They face an algorithmically curated field where past behaviours are selected at key commercial moments; the consumer's memory is now shaped not only by their experience but also by a commercial AI system.

2.1.2 | AI-Augmented Memory: The Construct

AI-AM is an externally co-produced memory mechanism through which algorithms record, retrieve and selectively resurface consumers' prior behavioural traces, such as previous searches and cart abandonments, at important decision moments.

AI-AM builds on earlier research on cognitive offloading (Sparrow et al. 2011; Clark and Chalmers 1998), human–computer interaction showing how algorithms shape what users delegate to automated support (Yuan et al. 2025) and psychological traits that shape reliance on AI in decision-making (Küper and Krämer 2025). AI-AM extends this literature by focusing on how platforms reactivate consumers' prior behavioural traces and the effects on sustainability, not just trust or reliance calibration alone. Unlike earlier external-memory tools such as shopping lists or bookmarks, AI-AM is commercially motivated, algorithmically selective and behaviourally predictive: the system does not simply store and retrieve, it actively prioritises which traces are resurfaced, when and in what persuasive context.

This study does not assess individual commercial platforms or their specific machine-learning architectures. It examines consumer-facing functions, such as trace-based recommendations and reminders, rather than their underlying computational architectures.

AI-AM has surface features similar to several adjacent constructs, so its boundaries should be clearly defined. Table 1 illustrates this directly, showing that several platform constructs

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TABLE 1 | Distinguishing AI-augmented memory from adjacent constructs.

Construct	Core function	Relation to AI-AM
Algorithmic personalisation	Tailors content, offers or search results to an individual user	AI-AM is a specific form of personalisation that additionally reactivates a stored prior behavioural trace
Recommendation systems	Generates ranked product or content suggestions	Only become an instance of AI-AM when a suggestion draws on and resurfaces a consumer's recorded trace, not aggregate similarity
Retargeting	Recontacts a consumer with content related to a prior interaction, typically through paid advertising	AI-AM may operate through this delivery channel, but it is not the mechanism itself
Digital nudging	Alters choice architecture (defaults, framing, sequencing) to steer decisions	AI-AM can supply the trace-based material that makes a nudge personally relevant, but nudging does not require memory reactivation
Ordinary external-memory aids (shopping lists, bookmarks)	Consumer-controlled storage and retrieval of information	Retrieval is initiated by the consumer on demand; AI-AM is platform-initiated and selectively reactivated without a concurrent request
Review summaries, chatbots and visual search	Support a specific decision task through synthesised or conversational information	Function as AI-AM only where they draw on and resurface the consumer's own recorded traces; otherwise, they are decision-support interfaces, not memory reactivation
Trust, perceived control and attitude constructs	Describe a consumer's psychological orientation toward an AI system	Plausible consequences of AI-AM exposure, not the mechanism itself

Note: The key factor across all comparisons is the same: whether the mechanism reactivates a consumer's own previously recorded behavioural trace, without a concurrent retrieval request and reintegrates it into an active decision episode. AI-AM is present only where this reactivation step occurs.

(discussed later in this paper) are AI-AM mechanisms only under specific conditions, not by default.

2.1.3 | The Sustainability Blind Spot in AI-AM Research

Despite this theoretical development, AI-AM's impact on sustainability has not been examined. The main study providing this paper's data (see Section 3.2) examined cognitive efficiency, decision quality and consumer experience. It is still unclear if AI-AM systematically biases consumption toward sustainable or unsustainable outcomes; addressing this blind spot in AI-AM's theory is the first part of the gap identified in Section 1.

2.2 | AI and Sustainable Consumption: A Double-Edged Relationship

2.2.1 | How AI Can Advance Sustainable Consumption

AI systems have genuine potential to support SDG12. The accuracy of personalisation can improve product–consumer fit, reducing trial-and-error purchasing that generates waste (von Zahn et al. 2025; Guha et al. 2021). Recommendation systems can surface durable, energy-efficient or lower-impact alternatives that consumers otherwise might not find (Satornino et al. 2024). AI-enabled review summarisation can help consumers process information faster and make digital purchase decisions more confidently (Li et al. 2025). Predictive analytics can align inventory and delivery with actual demand, reducing oversupply and packaging waste across the system (Guha et al. 2021; Satornino et al. 2024).

2.2.2 | How AI Can Undermine Sustainable Consumption

AI's abilities that support sustainable consumption can also reinforce unsustainable patterns. Personalisation focused on conversion usually recommends frequent, high-margin purchases rather than durable or low-impact ones (Bergianti 2025). Promotional strategies that create a sense of scarcity, such as flash sales and countdown timers, can trigger impulse buying online (Wu et al. 2021; Qu et al. 2023). Such purchases may also lead to post-purchase regret and product returns (Das and Kunja 2024; von Zahn et al. 2025). Another

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concern is that AI systems trained using engagement data may amplify unverified or exaggerated claims about sustainability (Guglyuvatyy 2026).

Importantly, the literature has not yet explained how AI mechanisms drive unsustainable consumption at the cognitive level. This paper presents AI-AM as a plausible candidate. By managing the memory environment where consumers recall, evaluate and choose, AI systems can shape the cognitive preconditions of sustainable and unsustainable consumption before any direct persuasion occurs.

2.2.3 | The Gap: A Cognitive Mechanism

Reviewing the AI-sustainability literature more broadly, three additional limitations reinforce the case for a memory-centred account. First, existing studies note the effects but do not explain the cognitive processes causing them (Bergianti 2025; Satornino et al. 2024). Second, most studies focus only on AI's positive or negative role instead of exploring when the same system triggers different pathways. Third, Asia-Pacific consumer contexts remain underrepresented (Jain et al. 2024). The AI-AM framework developed in Section 2.4 addresses these limitations by providing interpretive propositions instead of definitive tests. However, fully solving all three problems is a goal for the larger research this paper initiates, not something a single study can achieve.

2.3 | Contextual Conditions Shaping AI Influence in Consumer Behaviour

2.3.1 | Singapore as a Theoretically Productive Context

Singapore is a productive setting for AI-sustainability research. As one of the world's most digitally advanced consumer markets (International Trade Administration 2025), Singapore has a mobile penetration rate of 165.0% (Singapore Department of Statistics, n.d.). Shopee, Lazada and TikTok Shop, the leading e-commerce platforms in Singapore, are among the most algorithmically sophisticated retail AI systems in the Asia-Pacific region. Singapore's Hofstede cultural profile shows high collectivism, high power distance, low uncertainty avoidance and moderate restraint (Hofstede et al. 2010; Triandis 1995).

This profile provides a useful lens for interpreting the findings. National cultural scores are viewed as contextual framing rather than as individual-level predictors. High collectivism aligns with the finding that AI-curated social proof strongly persuades participants. High power distance may explain why several participants showed deference to AI recommendations. Although national scores indicate tolerance for ambiguity, many Singaporean consumers show significant risk-aversion in purchasing. This suggests that the Uncertainty Avoidance Index (UAI) operates differently at the individual level than Hofstede's national scores imply.

2.3.2 | The Autonomy Question

Consumer autonomy is especially important for responsible consumption in AI-mediated environments. SDG12 establishes the goal of responsible consumption and production, but sustainable behaviour is more likely to persist when consumers experience their choices as self-directed rather than externally steered (United Nations 2015; Webb et al. 2013; Schulze and Janssen 2024). Recent research identifies three distinct autonomy stances that consumers adopt toward AI influence: negotiated autonomy (conscious, selective engagement); delegated agency (outsourcing tasks while retaining authority); and invisible influence (unperceived AI-influenced decision-making) (Puntoni et al. 2021; Longoni and Cian 2022). This paper argues that an individual's autonomy stance determines which AI-AM sustainability pathway is activated. Specifically, invisible influence, where consumers cannot perceive AI's impact on their decisions, is the main concern for sustainability.

2.4 | Theoretical Framework: AI-AM as a Dual-Pathway Sustainability Gate

The three literature reviews above raise an unanswered question: What happens when a platform's reactivation of a consumer's behavioural trace meets a sustainability-related decision? This paper proposes that AI-AM sits at this intersection, acting as a dual-pathway sustainability gate within the EKB consumer decision journey. Figure 1 presents the conceptual model.

3 | Methodology

3.1 | Research Design and Epistemological Positioning

This study employs an interpretivist qualitative design (Guba and Lincoln 1994; Denzin and Lincoln 2011). The primary analytical method is secondary thematic analysis (Heaton 2008; Braun and Clarke 2008), which involves reanalysing an existing qualitative dataset through a new theoretical lens. Secondary analysis is justifiable on three grounds: the original data relate to the new question, the new question was not the original study's focus and the re-analysis provides new theoretical value.

3.2 | Primary Study Context

The dataset originates from a prior qualitative study examining AI-augmented memory and consumer decision-making in Singapore (Ng 2026). The primary study conducted 30 semi-structured interviews with active e-commerce consumers across Shopee, Lazada, TikTok Shop and Carousell. The ethical basis for secondary analysis is established by the original informed consent process, which authorised data retention and academic publication.

This paper differs from the prior study in four ways. First, the research question is distinct: where the prior study asked how AI-AM reshapes consumer decision-making, this study asks how AI-AM activates sustainable or unsustainable consumption pathways. Second, the analytical framework is entirely new: the sustainability coding framework, the dual-pathway model and the autonomy stance typology were not in the prior

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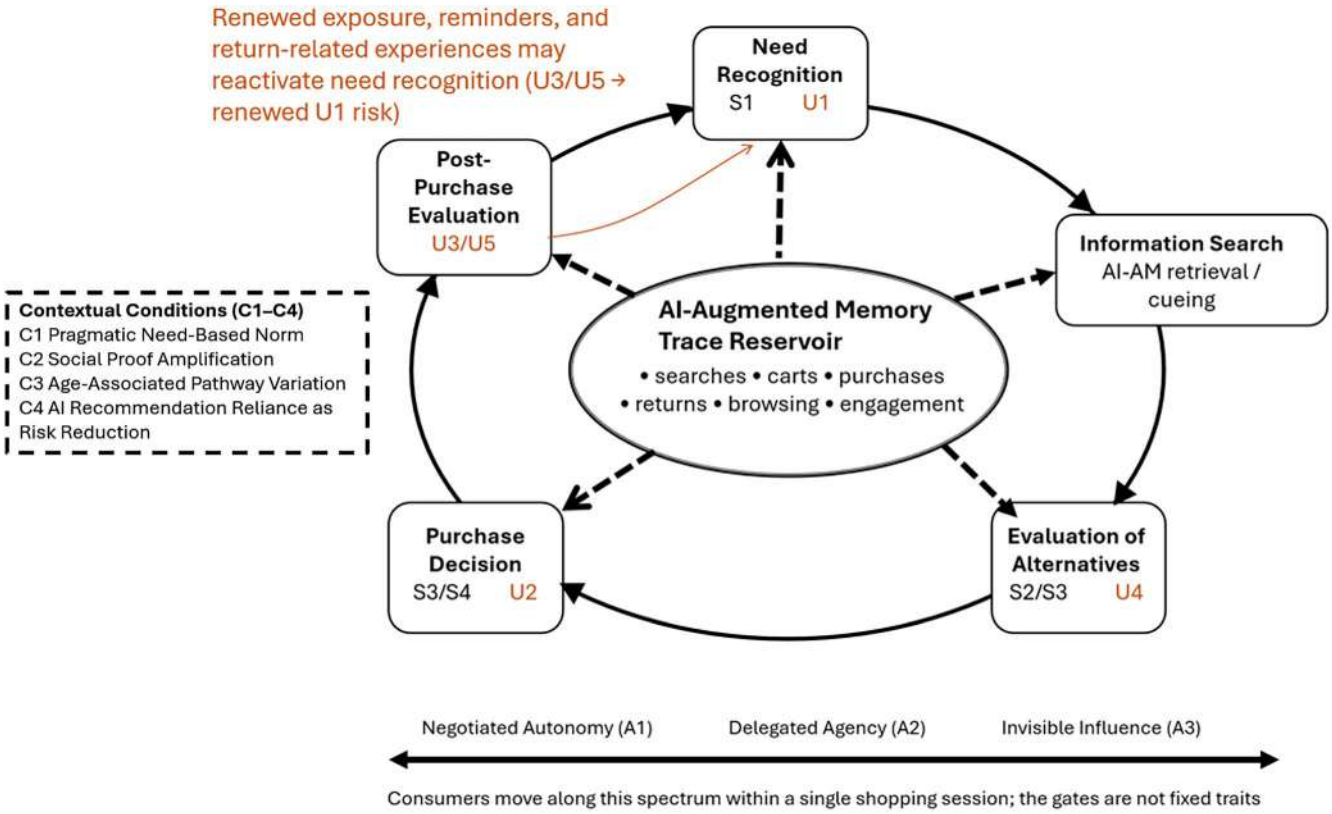


FIGURE 1 | AI-AM as a recursive dual-pathway sustainability gate within the EKB model. S = sustainable pathway code; U = unsustainable pathway code; A = autonomy stance; C = contextual condition. The AI-AM trace reservoir feeds all five stages of the consumer decision journey (dotted spokes). Relevant sustainable (S) and unsustainable (U) pathway codes are shown at the decision stages where they apply. The orange arrow indicates the specific possibility that renewed exposure, reminders and return-related experiences may reactivate need recognition and renew U1 risk, rather than the journey terminating at purchase. Contextual conditions (C1–C4) are shown as situated background conditions rather than as embedded causal moderators.

analysis. Third, the theoretical contribution is independent: EKB memory is reconceptualised as a sustainability gate, AI literacy is framed as a sustainability competence and the sustainability-aligned AI-AM design typology are unique to this paper. Fourth, the prior study did not develop or claim the contextual-conditions argument using Hofstede-based framing for pathway activation.

3.3 | Sampling Strategy and Participant Profiles

Participants were recruited using purposive theoretical sampling (Patton 2015) to ensure diversity in age, occupation, education, digital literacy and platform usage, with snowball sampling used in later stages. The sample of 30 participants (16 female, 14 male; ages 18–66; M = 46) exceeds the established thematic saturation thresholds for primary data collection (Guest et al. 2006). Because this paper is a secondary analysis of an existing fixed corpus rather than new data collection, the main concern is whether the existing dataset is theoretically or conceptually sufficient for the present analysis. Recent guidance suggests that this should be considered separately from data saturation (Lim 2026) (see Section 3.5). The sample size and diversity are appropriate for interpretive theory-building qualitative research. All interviews were conducted in Singapore, from April to May 2025.

3.4 | Data Collection

The researcher conducted all the interviews either through video conferencing or in person. These interviews lasted a total of 1303 min (approximately 171,000 words of verbatim transcribed data), with individual interviews ranging from 17 to 77 min and an average length of 43.4 min. The interview covered eight themes: recent purchases, search and discovery, memory and tracking, reviews and trust, post-purchase reflection, AI influence and autonomy, sustainability and future expectations. The question about sustainability was asked second to last, after discussing realistic accounts of AI-mediated purchasing, to capture sustainability-related behaviour while partially controlling for social desirability bias.

The interview protocol followed the EKB decision-stage structure and the developing AI-AM model. Although not solely focused on sustainability, relevant material emerged from both a specific sustainability question and participants' accounts of regret, returns, restraint, product fit and platform influence. Appendix A provides the full interview domains, verbatim example prompts and their relevance to the present secondary analysis.

This protocol was designed for AI-AM and decision-making broadly, not just for sustainability; therefore, this present analysis

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is a theoretically guided secondary interpretation of material that includes, but is not limited to, sustainability-related responses. This constraint is also noted as a limitation in Section 6.4.

3.5 | Secondary Analysis Procedure

The secondary analysis followed a theoretically driven variant of reflexive thematic analysis (Braun and Clarke 2008, 2019) adapted for secondary data (Heaton 2008). The procedure comprised eight steps:

1. full transcript immersion
2. sustainability-focused protocol review
3. provisional code generation (A1–A10)
4. codebook development and pilot (A11–A15)
5. full corpus coding (A1–A30)
6. a retrospective sufficiency check across the full coded corpus
7. negative case analysis
8. theoretical model construction

Step 6 investigated whether later interviews introduced any materially new codes or relationships. This is consistent with the distinction between data saturation (a stopping rule for data collection) and theoretical sufficiency (assessed retrospectively on an already fixed corpus) (Lim 2026). No new material was identified in pathway codes beyond A22 or in contextual-condition and autonomy codes beyond A27, indicating that the existing corpus offers sufficient conceptual grounding for this paper's claims.

In Step 7, instances where a participant's account did not match the expected pattern of behaviour were noted during coding; five such cases were subsequently documented and analysed for their consequences on the coding framework.

The sustainability coding framework comprised 16 codes divided into four groups: *sustainable pathway* (S1–S4), *unsustainable pathway* (U1–U5), *contextual conditions* (C1–C4) and *autonomy stance* (A1–A3). Each code was operationalised with a definition, inclusion/exclusion criteria, verbatim anchor quotations and EKB stage mapping (see Appendix A).

3.6 | Researcher Reflexivity

The researcher earned a Doctor of Business Administration during this research programme and is an adjunct marketing lecturer and active e-commerce practitioner across the platforms studied. This pracademic approach enhanced sensitivity to participants' accounts but introduced the risk of confirmatory interpretation. To address this, the researcher used negative case analysis, stayed closely grounded in participants' verbatim accounts and kept a reflexive memo during coding. As a secondary analysis, the study also analysed accounts not framed by participants in sustainability terms, which reduced social desirability bias but required careful interpretation of the findings.

3.7 | Trustworthiness

Credibility is supported by 21.7 h of interview data, comparison with the prior AI-AM interpretation of the same corpus and systematic negative case analysis (Lincoln and Guba 1985). Since this comparison uses the same interviews rather than a separate dataset, it is not triangulation in the strict sense. Transferability is supported by detailed descriptions of the Singapore context and participant profiles, with claims based on theoretical rather than statistical generalisation (Shenton 2004). Dependability is ensured by the detailed codebook with clear criteria and verbatim anchors. Confirmability is supported by an audit trail that includes the coding log, the codebook and the full transcripts. An external independent qualitative researcher conducted a structured review of the coding framework by blind coding three anonymised excerpts, answering three code-boundary questions and evaluating the conceptual model (Figure 1). Their independent codes matched the original codes for two excerpts; for the third, they assigned two codes (C4 and S3) instead of one, reflecting the excerpt's overlap. Two boundary questions were confirmed as meaningful, while the third (S1/U1 boundary) revealed real ambiguity due to AI recommendations based on participants' prior browsing history, now noted as a limitation (Section 6.4). Ethical approval followed Heaton (2008) and the original ethics protocol.

Section 6.5 outlines a future research agenda, such as quantitative validation, experimental design and cross-cultural replication, which builds on the theoretical model developed here and guides the next steps for this interpretive, theory-building contribution.

3.8 | Practitioner-Reflexive Account

To avoid the theoretical risk of a single perspective, this paper draws on the researcher's practitioner-reflexive account of multicategory retail operations in Singapore alongside consumer interviews. Although this account has not been independently audited or documented to the standard of the consumer-interview dataset, it offers interpretive context on algorithmic visibility, pricing and promotion strategies, return-cost management and platform-seller governance.

The practitioner-reflexive account provides interpretive context on how platform incentives may contribute to promotional escalation, shifting return costs and seller-side path dependence. It contextualises rather than independently verifies the consumer findings. This practitioner-reflexive data layer (Van de Ven 2007; Hine 2015) offers an interpretive view of the behind-the-scenes commercial architecture where consumer-facing AI-AM mechanisms operate. Consumer interviews illuminate how AI-AM shapes buyer decision-making; however, the practitioner account offers a reflexive perspective on *why* these mechanisms are designed as they are. This shows which platform incentives drive promotional escalation, how return-cost structures are constructed and modified and the consequences of AI-mediated path dependence for sellers.

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This dual-perspective approach aligns with pracademic research design (Boyer 1990; Hollweck et al. 2022), which explicitly integrates practitioner knowledge as a legitimate source of theoretical insight. Section 4.6 uses the practitioner observations to interpretively contextualise the consumer findings, not to independently verify or replace them. Practitioner claims are clearly distinguished from consumer-interview evidence in the findings.

4 | Findings

The narrative findings are organised around the dual-pathway structure, followed by the contextual-condition and autonomy stance findings and summarised in a three-pattern model. Table 2 summarises this evidence for all 16 codes, showing participant accounts, whether sustainability relevance was explicit or inferred, a representative quote and the corresponding EKB stage for each code.

4.1 | The Sustainable Pathway: How AI-AM Can Support SDG12 (S1–S4)

Four mechanisms through which AI-AM activates a sustainable consumption pathway were identified. These mechanisms share a common logic: AI-AM supports sustainable outcomes when it confirms rather than manufactures consumer need, improves rather than accelerates purchase decisions and preserves rather than erodes deliberative space.

It is important to first clarify that some mechanisms described below, including review summarisation, specification filtering and cross-platform price comparison, are AI-enabled decision-support functions, not AI-AM as strictly defined in Table 1. They are included here because participants experienced them as operating alongside or facilitated by AI-AM trace reactivation. For example, a pre-purchase summary that surfaces because the platform has retained the consumer's prior search or cart activity, rather than being a generic, non-personalised function. Any quotation describing a hoped-for feature rather than an experienced one is noted below.

4.1.1 | S1: Need-Confirmation Cueing

The most frequently cited sustainable mechanism involved AI-AM resurfacing a prior product trace that confirmed a genuine, pre-existing consumer need. In these accounts, the AI cue did not create the need but confirmed and activated it, functioning as an externally managed memory aid for decisions that were already internally motivated.

> I purchased based on needs, okay? Usually things that I will need, yes. And even for clothes, I have to find the design I like, and I don't see myself buying ten pieces a month.
>
> (A1, Female, 45, Teacher)

> I always tell myself: you need it. If I cannot answer that question, I don't buy. Very simple. You can waste some money, yeah, but that question saves more than it costs.
>
> (A24, Male, 63, Property Maintenance)

A2, an engineer with high platform literacy, offered a self-aware account of need-confirmation as a *deliberate counter-strategy* against AI-triggered impulses:

> I know AI's role is actually to help the seller sell more. So as a discerning buyer, I know I cannot just take everything wholesale. I have to know what I really, really want, and then use AI as a tool to find it more efficiently.
>
> (A2, Male, 49, Engineer)

4.1.2 | S2: Accuracy-Enabled Waste Reduction

Eleven participants described AI-enabled decision-support functions, specifically review summarisation, specification filtering and comparison compression, sometimes operating alongside AI-AM trace reactivation, as improving the accuracy of product selection in ways that reduced returns, avoided unused purchases and generated less consumption waste.

> The review summary tells me whether the size is true, whether the material is see-through, whether it lasts. Because although yes, you can have free return, I still find it troublesome to return. I want to fit myself as far as possible.
>
> (A1, Female, 45, Teacher)

> They should suggest, before you click purchase, a single-page summary: you are buying this, in this colour, this size. Please review them again. That would provide a lot of joy, and save a lot of waste.
>
> (A24, Male, 63, Property Maintenance)

4.1.3 | S3: Budget-Disciplined Curation

Nine participants described platform-supported budgeting and comparison tools, such as voucher stacking and cross-platform comparisons (sometimes linked to AI-AM reminders or retained shopping traces), as forms of financial self-governance that helped control their spending.

> I would like AI to help me compare across platforms. Now we have Shopee, Lazada, Carousell, if AI can compare the price, I think that would be even better. Then you don't overpay, and you don't buy twice.
>
> (A1, Female, 45, Teacher, describing a hoped-for feature, not a function she currently experiences)

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TABLE 2 | Sustainability evidence summary.

Code	Name	Pathway	*N*	Explicit/inferred	Representative quote (truncated)	EKB stage
S1	Need-confirmation cueing	Sustainable	9	Explicit	I always tell myself: you need it. If I cannot answer that, I don't buy. (A24)	Need recognition
S2	Accuracy-enabled waste reduction	Sustainable	11	Explicit (8) inferred (3)	I will end up getting an item I can use for long, instead of buying things I chuck aside. (A29)	Evaluation
S3	Budget-disciplined curation	Sustainable	9	Explicit (6) inferred (3)	I spread it out: fifty dollars this month, fifty next. The platform is my budget diary. (A3)	Evaluation; purchase
S4	Deliberative pause	Sustainable	10	Explicit (7) inferred (3)	The fact it's still in the cart means I'm not really looking for it. (A1)	Purchase decision
U1	AI-triggered impulse purchase	Unsustainable	11	Explicit	I end up buying things I don't need, just because they pop up. (A16)	Need recognition
U2	Promotional escalation	Unsustainable	11	Explicit	The platform knows when I weakened last time. It uses that. (A9)	Purchase decision
U3	Returns-related waste	Unsustainable	10	Explicit (6) inferred (4)	A lot of plastics and cardboard. I feel there is a lot of wasted material. (A24)	Post-purchase
U4	Category lock-in	Unsustainable	7	Explicit (5) inferred (2)	AI never highlights that whatever they present is in the interest of sustainability. (A24)	Evaluation
U5	Overconsumption normalisation	Unsustainable	7	Inferred (dominant)	AI makes it so easy to buy more than you need, and that is the problem. (A2)	All stages
C1	Pragmatic need-based norm	Contextual condition	9	Inferred	If AI shows me something I don't need, I scroll past it. (A1)	Cross-stage
C2	Social proof amplification	Contextual condition	7	Inferred	The crowd cannot be wrong. Unless the crowd is fake. (A23)	Evaluation; purchase
C3	Age-associated pathway variation	Contextual condition	8	Inferred	I always tell myself: you need it. If I cannot answer that question, I don't buy. (A24)	Cross-stage
C4	AI recommendation reliance as risk reduction	Contextual condition	7	Inferred	You have more choices. You don't have to spend a lot of time looking for it. (A5)	Evaluation; purchase
A1	Negotiated autonomy	Autonomy	10	Explicit (8) inferred (2)	My understanding of how this technology works, that is my protection. (A2)	Key cross-stage autonomy condition
A2	Delegated agency	Autonomy	8	Inferred	The AI does the wide search. I do the final decision. (A1)	Info Search; evaluation
A3	Invisible influence	Autonomy	7	Inferred (primary)	Maybe subconsciously I have used those suggestions. I don't consciously use them. (A30)	Cross-stage (covert)

Note: N = number of participants contributing evidence. 'Explicit' = participant directly addressed sustainability in connection with the behaviour. 'Inferred' = sustainability significance attributed through secondary coding. C and A codes function as modifiers alongside S/U codes rather than independent pathway codes.

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> I spread it out. Okay, this month maybe I spend fifty dollars, next month another fifty. Unless it's very urgent. So I use the platform to track that, it's my budget diary.
>
> (A3, Male, 54, Ex-Event Manager)

4.1.4 | S4: Restraint Through Deliberative Pause

Ten participants described the shopping cart or wishlist as a deliberative holding space, a temporal buffer between exposure and purchase. AI-AM's cart-reminder mechanism paradoxically supported consumption restraint by making the decision not to purchase explicit rather than passive.

> The fact that it has been left there means that I am actually not really looking for it. So I ignore the reminder. Sometimes the reminder is the signal that I don't need it.
>
> (A1, Female, 45, Teacher)

> Now I just leave it there. I say, I will come back to it later and decide whether I want it. And a lot of the time, I don't go back. The cart is my cooling-off room.
>
> (A16, Female, 47, Accountant)

4.2 | The Unsustainable Pathway: How AI-AM Undermines SDG12 (U1–U5)

Five mechanisms through which AI-AM activates an unsustainable consumption pathway were identified. Unlike the sustainable pathway, where AI-AM confirmed consumer intent, the unsustainable pathway was characterised by AI-AM manufacturing or escalating consumption impulses beyond genuine need.

4.2.1 | U1: AI-Triggered Impulse Purchase

Eleven participants described purchases that originated from an AI recommendation, notification or feed cue for a product they had not planned to buy. These accounts differ from S1 because of the absence of prior intent: the AI cue was the primary trigger rather than a confirmation of existing motivation. This distinction is conceptually clear but can be difficult to apply because AI recommendations are frequently generated from the participant's browsing history, which can be read as evidence of a pre-existing, if unarticulated, need, particularly when prior intent was weak or only partially formed before the cue appeared. An independent review of the coding framework (Section 3.7) identified this as a genuine boundary ambiguity rather than a coding error.

> Sometimes I end up buying things that I don't need, just because I was searching for something and other things pop up. The AI shows me things and I think, oh, maybe I need that too. And then I buy it. And then I wonder why.
>
> (A16, Female, 47, Accountant)

> I have friends who get very addicted by TikTok live streams. They always have special offers, click now, click now. And the AI thinks: okay, this person is watching, this person is interested. And then the recommendations keep coming. The seduction is relentless.
>
> (A9, Female, 50+, Business Owner)

The TikTok Shop context warrants specific theoretical attention because it has a qualitatively different AI-AM architecture than search-based platforms. On Shopee or Lazada, AI-AM is mainly utilitarian: consumers arrive with a category or product in mind and AI resurfaces prior traces to confirm or extend their search. In contrast, on TikTok Shop, AI-AM is hedonistic and entertainment-focused (Luo et al. 2025; Wongkitrungrueng and Assarut 2020): consumers arrive to watch, not to shop and purchase intent is generated through parasocial engagement rather than prior product searches (Lo et al. 2022; Qu et al. 2023). Participants' accounts and prior research suggest these platforms may view passive engagement (e.g., pause behaviour and scroll-back) as indicators of purchase interest without explicit signals from the consumer. This results in what this paper terms *hedonic AI-AM*: a cueing architecture that appears to generate purchase intent from entertainment engagement, not through confirming a pre-existing intent. Where utilitarian AI-AM (S1) asks 'did you want this before?', hedonic AI-AM (U1) asserts 'you want this now'.

This distinction may help contextualise the age-associated pattern reported under C3 below. TikTok Shop users in this sample showed higher AI-triggered impulse purchases (U1) regardless of age, showing that platform design matters more than demographics. A9's account of friends becoming absorbed in livestream commerce illustrates a plausible feedback loop: the AI interprets watching as interest, interest as intent and intent as licence to intensify recommendations, continuously inferring purchase intent from attention data rather than from explicit signals. This is the clearest example of U1 in the dataset, linked to the content-commerce fusion design (Lo et al. 2022) rather than consumer characteristics, aligning with evidence that livestream platforms prompt impulsive purchase behaviour across consumer groups through design, rather than individual differences (Xu et al. 2020).

4.2.2 | U2: Promotional Escalation and Flash-Sale Trapping

Eleven participants described experiences in which AI-AM mechanisms, such as promotion-linked notifications, countdown timers and event-based campaigns (e.g., 11.11 and 9.9), escalated purchasing beyond the amount intended. The mechanism combined AI-AM's memory resurfacing capability with artificial urgency, bypassing deliberative restraint.

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> My husband bought a Philips iron at six hundred plus. Then that night he checked again, price had dropped to four hundred plus. He was very mad. Then the next day it dropped to three hundred plus. Promotion really makes anyone very busy in that way.
>
> (A1, Female, 45, Teacher)

> It has got to do with your willpower versus AI seduction. When the offer is within what I think is good for me, I click and I buy. The platform knows when I weakened last time. It uses that.
>
> (A9, Female, 50+, Business Owner)

4.2.3 | U3: Returns-Related Waste

Ten participants described return experiences illustrating a structural unsustainability created by AI-enabled frictionless purchasing: the ease of returns leads to more impulsive buying, which generates associated packaging waste.

> A lot of plastics and cardboard that will not have to be thrown away, I feel there is a lot of wasted material in all this online thing. I save time, yes, and it came on time, but then the packaging hits me.
>
> (A24, Male, 63, Property Maintenance)

> Some people, they get free returns, but they don't bother to return. They end up give away or just throw away all this thing. It is not sustainable, especially for the environment.
>
> (A23, Female, 44, Housewife)

4.2.4 | U4: Algorithmic Category Lock-In

Seven participants described a structural AI-AM failure where recommendation systems stopped updating consumer preferences, continued surfacing rejected brands and never surfaced eco-friendly or lower-waste product alternatives. This is significant because it identifies that a lack of AI-sustainability causes unsustainable consumption, a design failure independent of commercial incentives.

> The AI stopped learning at a point. It keeps just giving you what it thinks is the best option. I know that brand, I tried it once, it was terrible. But AI will still keep pushing it to me. It has done its job once. Then it just keeps repeating it.
>
> (A13, Female, 52, Marketing)

> AI never highlights that whatever they present is in the interest of sustainability. Their main focus is just to sell you as much as they can. Sustainability is way down in their priority list.
>
> (A24, Male, 63, Property Maintenance)

4.2.5 | U5: Overconsumption Normalisation

Seven participants described habitual over-purchasing patterns they attributed to the structural design of AI-mediated retail environments, which they experienced as normalising high-frequency, low-deliberation buying.

> People are buying more now, to the point where they're buying things that they don't need. And that actually has a negative impact on sustainability in the environmental sense. Gigantic warehouses. An army of delivery motorcycles. AI decision-making basically makes things so easy, and that is the problem.
>
> (A2, Male, 49, Engineer)

> I will openly admit that I am not a sustainable person. I think that if you're on Lazada and Shopee at all, you end up just buying a lot of things you don't need.
>
> (A13, Female, 52, Marketing)

4.3 | Contextual Conditions: Singapore's Setting and Pathway Activation (C1–C4)

Four contextual conditions consistently influenced which AI-AM pathway was activated. These conditions are presented as interpretive patterns, not as causal claims. In a single-country qualitative study, cultural inference is provisional and indicative. With that caveat, the following patterns were consistent enough across participants to serve as working propositions for future cross-cultural research.

4.3.1 | C1: Pragmatic Need-Based Norm

The most consistent pattern across participants, regardless of their income, was the cultural value of purchasing only what is genuinely needed. This norm was the pattern most closely associated with S1 (need-confirmation) and S4 (deliberative pause) over U1 (impulse purchase). This is consistent with Singapore's high long-term orientation (Hofstede et al. 2010).

> I just buy what I need. Why would I buy more? My friends keep buying things they don't use. That doesn't make sense to me. If AI shows me something I don't need, I scroll past it.
>
> (A1, Female, 45, Teacher)

4.3.2 | C2: Social Proof Amplification

Seven participants overly relied on AI-curated social signals, star ratings, review counts and 'customers also bought' indicators. In Singapore's collectivist culture, social consensus helped reduce uncertainty and legitimise choices for these participants. C2 affects sustainability both ways: it amplifies the sustainable

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pathway when signals favour durable products but amplifies the unsustainable pathway when social proof endorses promotional or unnecessary items.

> So many people say that is good, then I'll buy. I don't need to verify further. The crowd cannot be wrong. Unless the crowd is fake, and then I have a problem.
>
> (A23, Female, 44, Housewife)

4.3.3 | C3: Age-Associated Pathway Variation

Younger participants (18–21) more often described AI-triggered impulse purchasing, whereas older participants (63–66) more often described deliberative resistance. This age-associated pattern was not universal, however and platform type appeared more salient than age: TikTok Shop users reported higher U1 and U2 experiences across age groups, suggesting that platform design may shape the observed age-related variation.

> For me, I need to understand the fundamentals of what I am buying. I will research on YouTube, read independent reviews, move items from a to-do list to a shopping list before they ever reach my cart. The AI can help me find, but I decide.
>
> (A25, Male, 48, MBA Director)

4.3.4 | C4: AI Recommendation Reliance as Risk Reduction

Seven participants used AI recommendations as a primary risk-reduction mechanism.

> Actually, I think it's good, because then you have more choices. You don't have to spend a lot of time looking for it. So we just go for the lowest price, see which one is better. (A5, Male, 63, Bachelor's Degree, Transport Manager)

This example frames recommendations as a way to save time and reduce risk. This pattern contrasts with Singapore's low national uncertainty avoidance classification, underscoring the limitations of applying national cultural indices directly to individual consumer behaviour (Hofstede et al. 2010). AI recommendation reliance (C4) may relate to either pathway: when AI deference produced accurate fit, it was linked to S2; when it produced uncritical acceptance, it was linked to U1 and U3.

4.4 | Autonomy Stance: The Individual-Level Mechanism (A1–A3)

Autonomy stance emerged as the *primary individual-level mechanism* determining which pathway is activated. Three distinct stances were identified: negotiated autonomy (A1), delegated agency (A2) and invisible influence (A3). These stances are not fixed consumer characteristics but depend on the context and the product.

4.4.1 | A1: Negotiated Autonomy

Ten participants exhibited negotiated autonomy, consciously and selectively engaging with AI-AM cues. Negotiated autonomy was the strongest predictor of sustainable pathway outcomes. This suggests that consumer AI literacy (i.e., the capacity to identify and resist AI influence) is the key individual-level mechanism for activating sustainable consumption through AI-mediated channels.

> AI to me is a tool. I don't let it run my life. I don't just take everything that AI recommends as wholesale truth. My understanding of how this technology works, that is my protection.
>
> (A2, Male, 49, Engineer)

> It has got to do with your willpower versus AI seduction. When I know the AI is pushing something at me, I ask myself: would I have looked for this if the platform hadn't shown it? If the answer is no, I put it back.
>
> (A9, Female, 50+, Business Owner)

4.4.2 | A2: Delegated Agency

Eight participants described consciously outsourcing specific decision-making tasks (e.g., comparison and shortlisting) to AI-AM while retaining final purchase authority. The sustainability implications of delegated agency depend on the AI system's quality: delegation to accuracy-prioritising systems (S2) supports sustainable outcomes; delegation to conversion-optimised systems (U1, U2) can inadvertently activate the unsustainable pathway.

> Since they are able to generate a lot more examples, why not? I have more choices. The AI does the wide search. I do the final decision.
>
> (A1, Female, 45, Teacher)

4.4.3 | A3: Invisible Influence

Seven participants exhibited invisible influence, denying or underestimating AI's impact on their decisions; yet their interviews clearly showed that AI-AM mechanisms significantly influenced their purchasing. This stance has the most consequences for sustainability because it prevents the deliberative processes (e.g., need-questioning and comparison) that activate the sustainable pathway. The normalisation of AI-mediated retail has made its impact invisible, even to reflective and articulate consumers.

TikTok Shop's FYP (For You page) is an ideal environment for invisible influence, unlike search-based interfaces where AI-AM mechanisms are partly visible because consumers type a

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query and can see the cueing relationship between input and output. In the FYP design there is no query; consumers simply watch while the algorithm continuously reconfigures content in response to micro-engagement signals that they do not consciously provide (Bucher 2018). This makes the AI's influence invisible (Tufekci 2017), not because the consumer is inattentive, but because the platform hides the algorithmic layer within entertainment content, instead of showing it as a clear commercial recommendation system. This suppresses the conditions for negotiated autonomy (A1), which requires awareness of AI influence to allow negotiation. This platform-level invisible influence strongly supports the transparent design of AI (typology Principle 7) and AI literacy interventions that teach consumers to recognise hedonic AI-AM cueing as a distinct type of influence.

> The AI? I do not feel that it has influenced me. I just buy what I need. But... I mean, if I see it appearing, I look at it. And if it suits me, if I need it, I buy. So I suppose it does give me more options. But I decide.
>
> (A1, Female, 45, Teacher)

> Maybe subconsciously, I have used those AI suggestions as a base to make my purchase. I don't consciously use them. But they are there. And maybe I am shaped by them without knowing.
>
> (A30, Male, 47, Education Administrator)

4.5 | Synthesis: Three Cross-Cutting Patterns

Three cross-cutting patterns consolidate the paper's three core contributions. First, negotiated autonomy (A1) was the most consistent indicator of a sustainable pathway and invisible influence (A3) was the most consistent indicator of an unsustainable pathway (Ng et al. 2021; Long and Magerko 2020). Second, the interaction between the platform and the pathway is key. Accounts involving TikTok Shop more frequently featured U1 and U5 codes across all age groups, showing AI-AM sustainability outcomes are shaped by platform design, not only by consumer characteristics. Third, no participant recalled being shown sustainable alternatives without actively searching for them. This pattern suggests that sustainability signals were not prominent in routine platform curation. This universal gap drives design-level unsustainable consumption independent of consumer intent. The pattern is now clear: TikTok Shop uses a hedonic AI-AM architecture, where purchase intent is inferred from passive entertainment, rather than active search. This is architecturally different from utilitarian AI-AM (Shopee, Lazada), where AI-AM responds to consumer intent. Hedonic AI-AM seems to create intent directly from attention signals, skipping or shortening the deliberative stage. This distinction has direct design implications (typology Principles 4 and 5) and identifies platform architecture as the primary structural variable in the dual-pathway model.

4.6 | Practitioner Perspective: The Seller-Side Architecture of the Dual Pathway

The practitioner's reflexive account provides seller-side context for four pathway codes, viewed as interpretation rather than verified evidence. For promotional escalation (U2), new listings often faced pressure to join launch discounts for early algorithm visibility; once momentum started, algorithmic ranking boosted it only while promotions continued and search ranking usually dropped when sellers withdrew. This creates what the practitioner calls *algorithmic path dependence* (Arthur 1994): sellers do not just choose to run flash sales but also respond to incentives that make withdrawal costly. The practitioner describes this mechanism as *platform-engineered seller dependency*, which is underspecified in AI-sustainability research. Major shopping events (e.g., 11.11 and 9.9) are the most intense examples, with participation nearly unavoidable during peak traffic.

On returns-related waste (U3), platform 'change of mind' return policies, such as Shopee's 15-day window (Shopee 2025a), are commercial tools rather than government-mandated rights, meant to reduce friction and accelerate buying. Two-way returns, in the practitioner's experience, add a recurring cost per transaction; Shopee initially subsidised this before phasing out the Seller Shipping Fee Subsidy from 1 January 2026 (Shopee 2025b), transferring the cost to sellers. The practitioner calls this *platform decay* (Cutolo and Kenney 2021): platforms attract participation under favourable conditions, then progressively shift costs to sellers as their market position consolidates. This suggests that U3 reflects governance choices, not just promotional cueing and low-deliberation purchasing is rewarded when returns cost buyers nothing.

On algorithmic category lock-in (U4), in the practitioner's experience, dashboards appear to be designed around profit rather than product quality or environmental impact. Furthermore, an imitation-loop dynamic, where competitors rapidly replicate visible listings, narrows the viable product mix, discouraging durable, lower-frequency alternatives without explicit censorship.

Conversely, on Carousell, where direct buyer–seller interaction is embedded, the practitioner describes a trust dividend: buyers who consult on compatibility and authenticity before purchase appear more willing to pay a premium and less likely to return items, mirroring the S1 and S4 mechanisms by reinstating deliberation as a transactional requirement.

The practitioner agrees that platform architecture, not just consumer characteristics, is the key structural factor in the dual-pathway model: the unsustainable pathway reflects design choices favouring conversion volume over deliberation quality, while consumer AI literacy (negotiated autonomy, A1) is the main individual-level limit within that environment. However, platform changes may be needed for it to work fully, which is an interpretive proposition rather than a tested claim.

5 | Discussion

The paper makes three connected theoretical contributions. The first is redefining AI-AM as a dual-pathway sustainability gate in the EKB consumer decision journey. Contributions 2 and 3 are developed in Sections 5.2 and 5.3, with cultural findings used as contextual evidence throughout.

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5.1 | Contribution 1: AI-AM as a Dual-Pathway Sustainability Gate

The paper shows that AI-AM affects each EKB stage differently to create sustainability outcomes. At the need recognition stage, AI-AM can confirm an existing need (S1) or create a new one (U1). At the evaluation of alternatives stage, comparison compression can either improve product fit (S2, S3) or entrench consumers in a category loop that excludes sustainability-relevant options (U4). At the purchase decision stage, both the deliberative pause mechanism (S4) and promotional escalation (U2) represent AI-AM activating restraint or urgency through the same temporal cueing capability. At the post-purchase stage, returns-related waste (U3) shows the total sustainability cost of earlier AI-AM decisions.

The paper introduces *autonomy stance* as the individual-level mechanism determining which pathway is activated at each stage of the EKB model. This expands on Puntoni et al.'s (2021) experiential framework and links the hedonic AI contexts examined by Longoni and Cian (2022) to sustainability outcomes. Invisible influence (A3) has the greatest impact on sustainability because it prevents deliberative resistance. This introduces a new idea to the consumer vulnerability literature (Baker et al. 2005; Hermann et al. 2024): the findings position autonomy stance as an individual-level boundary condition associated with pathway activation. They suggest a form of contextual vulnerability in which platform design can obscure AI-AM's role in shaping choice, limiting consumers' capacity for deliberative resistance.

The cultural findings enrich Contributions 1 and 3 instead of standing alone as a separate claim. In the Singapore sample, the pragmatic need-based norm (C1) acted as a consumer-level sustainability competence supporting Contribution 3. It showed that participants with strong need-based norms were more likely to exhibit negotiated autonomy. Social proof amplification (C2), in contrast, acted as a collectivist amplifier of AI influence, activating either pathway depending on which aggregated signal it supported. These patterns are suggested propositions for future cross-cultural testing, not confirmed as replicable moderating effects (Jain et al. 2024).

5.2 | Contribution 2: AI-AM as an Unacknowledged Sustainability Architecture (EKB Memory as Contested Node)

The paper's second contribution is that AI-AM is an unacknowledged sustainability architecture: a commercially configured cognitive environment that shapes sustainable consumption before any explicit purchasing decision is made. This extends the literature (Thaler and Sunstein 2008) in two ways. First, conventional nudge architectures act at the moment of choice. AI-AM, however, acts earlier by shaping the memory environment before a choice occurs. Second, policymakers usually design conventional nudges to benefit consumers. In many commercial digital-retail settings, AI-enabled systems are usually optimised for engagement and conversion, making sustainability outcomes a by-product rather than an explicit design goal.

5.3 | Contribution 3: AI Literacy as the Key Sustainability Boundary Condition

The paper's third contribution is specifying consumer AI literacy as the capacity to recognise and negotiate AI-AM influence as a sustainability competence rather than just a digital skill. AI literacy has been defined as a multi-dimensional competency encompassing awareness of AI capabilities, critical evaluation of AI outputs and the ability to engage with AI systems as informed users (Ng et al. 2021; Long and Magerko 2020). This paper extends that definition to consumer contexts: the key capacity is not general platform knowledge but the practised skill to perceive AI-AM influence during a purchasing decision and to resist or redirect it. This distinguishes negotiated autonomy from invisible influence. The finding that negotiated autonomy (A1) was the pattern most consistently linked to sustainable pathway activation provides a previously undocumented connection between AI literacy and sustainable consumption. This connection is clear: AI literacy research identifies critical evaluation and autonomous engagement as core competency components (Ng et al. 2021). When applied to consumption, these components distinguish negotiated autonomy from invisible influence. Several participants partially moved away from invisible influence during the interview. This suggests that raising consumer awareness of AI-AM mechanisms could be a low-cost, high-impact way to promote sustainability.

5.4 | The Sustainability-Aligned AI-AM Design Typology

Table 3 presents the sustainability-aligned AI-AM design typology, a framework for platform designers and policymakers, based on the dual-pathway findings. The typology is anchored in three regulatory and policy frameworks: the European Green Deal's product sustainability requirements (European Commission 2019), the EU Artificial Intelligence Act's provisions on deceptive commercial AI practices (European Parliament and Council of the European Union 2024) and Singapore's Personal Data Protection Commission (PDPC) guidelines on AI recommendation and decision systems (PDPC 2024). The typology comprises eight design principles across three categories: sustainability-enhancing AI-AM (four principles), unsustainability-reducing AI-AM (two principles) and autonomy-preserving AI-AM (two principles).

The typology's relevance to live-stream commerce platforms such as TikTok Shop deserves comment, as this is the most challenging design environment for sustainability-aligned AI-AM. Principle 4 (temporal commitment architecture) cannot involve a shopping-cart cooling-off period because the purchase pipeline is embedded within real-time entertainment and flash offers usually expire during the stream. A platform-appropriate implementation would be a 'purchase pause' feature: a brief, opt-in prompt that appears when a viewer taps 'buy now', asking 'You've been watching for X minutes. Still want to buy?' with a five-second delay before checkout completes, preserving stream flow while inserting the deliberative moment that hedonic AI-AM removes. Principle 5 (promotional friction injection) would manifest as a price anchor displayed alongside the live offer, showing regular price,

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TABLE 3 | Sustainability-aligned AI-AM design typology (with evidence basis).

Category	Design principle	Implementation specification	Codes addressed	Implications	Evidence basis
Sustainability-enhancing	Need-confirmation architecture	Resurface only items matching stated need profile; gate recommendations against owned items.	S1, S4	Require opt-in for promotional cueing separate from need-based reminders. Transparency obligations on reminder trigger configuration.	Empirically grounded: S1 (A1, A24, A29); S4 (A1, A16, A24).
Sustainability-enhancing	Accuracy-first review curation	Surface review content prioritising fit, durability and specification accuracy; flag product-description discrepancies.	S2, U3 prevention	Penalise sellers with systematic return patterns. Mandate return-rate data display.	Empirically grounded: S2 (A1, A24, A26, A29); AI-AM component applies per Table 1 scope note.
Sustainability-enhancing	Cross-platform budget comparison	Provide price-history transparency and cross-platform anchoring; enable consumer-controlled budget ceilings.	S3	Open price-history APIs. Require price anchoring transparency in promotional events.	Empirically grounded: S3 (A1, A3, A26, A29); AI-AM component applies per Table 1 scope note.
Sustainability-enabling pause	Temporal commitment architecture	Build mandatory review intervals for high-frequency purchases; enable consumer-set cooling-off rules.	S4, U1 prevention	Opt-in deliberation tools as standard feature. Consider mandatory cooling-off for purchases above defined thresholds.	Empirically grounded: S4 (A1, A16, A24); cart-as-pause described explicitly by multiple participants.
Unsustainability-reducing	Promotional friction injection	Introduce a deliberative prompt before flash-sale purchase: display last-viewed price and days since cart addition.	U2 reduction	Potential relevance where a platform practice meets the statutory threshold for prohibited manipulative or deceptive AI techniques; ordinary countdown mechanisms are not presumptively within Article 5.	Analytical inference: U2 (A1, A9, A10); extrapolated from negative case A10 (learned to resist).
Unsustainability-reducing	Sustainability signal integration	Surface energy-efficiency ratings, durability scores, packaging waste indicators as standard product attributes.	U4, U5	Participate in standardised sustainability labelling. Consider minimum sustainability-disclosure standards for AI-curated product surfaces, aligned with the European Green Deal objectives.	Analytical inference: U4 (A13, A24, A30); no participant reported seeing sustainability signals unprompted. Normative.

(Continues)

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TABLE 3 | (Continued)

Category	Design principle	Implementation specification	Codes addressed	Implications	Evidence basis
Autonomy-preserving	AI influence transparency	Disclose when a product is AI-recommended vs. organically retrieved; allow consumers to toggle AI-AM mechanisms.	A1 enabling, A3 preventing	Transparency controls as standard. Singapore PDPC 2024 AI advisory guidelines on explainability.	Empirically grounded: A3 (A1, A12, A30) establish AI influence is invisible. Normative.
Autonomy-preserving	AI literacy prompts	Insert periodic plain-language explanations of how AI-AM recommendations are generated; prompt quarterly settings review.	A3 reducing	Singapore Smart Nation initiative (Smart Nation Singapore 2023); national AI literacy programmes should include consumer AI-AM awareness as a specific competency.	Analytical inference: A30's partial emergence from Invisible Influence during interview. Normative.

Note: Evidence Basis column distinguishes: empirically grounded = participant accounts directly support the design principle; analytical inference = principle derived from pattern in data; normative = design or policy recommendation extending beyond data to actionable guidance. EU AI Act references to final adopted text. Singapore PDPC references to 2024 AI advisory guidelines. SDG12 targets: 12.1 (sustainable consumption); 12.5 (waste reduction); 12.8 (awareness).

last-purchased price and purchases of similar items in the past 30 days. Together these address a failure of live-stream commerce in the way it condenses of need recognition, evaluation and purchase decision into a single moment of hedonic engagement, without requiring the viewer to exit the entertainment context. Both implementations need platform co-operation and are informed by the broader concerns in Article 5 of the EU AI Act regarding manipulative and opaque AI-mediated influence, though their applicability depends on the statutory threshold and case details (European Parliament and Council of the European Union 2024).

The practitioner account in Section 4.6 offers additional reflexive context to the consumer data for three typology principles. Principle 4 (temporal commitment architecture) aligns with the practitioner's view that without a deliberative pause in return-cost design, the frictionless environment the typology recommends counteracting is created. Principle 5 (promotional friction injection) is informed by the practitioner's account of seller-side algorithmic path dependence: platforms requiring promotional participation for visibility cannot straightforwardly claim that consumers freely choose to engage in promotions. Principle 6 (sustainability signal integration) is reinforced by the imitation-loop finding: environments that reward sales velocity over differentiation will not allow sustainability-labelled alternatives to emerge organically without design intervention. This strengthens the typology's practical plausibility, though it is not independently verified evidence.

6 | Implications, Limitations and Conclusion

6.1 | Theoretical Implications

This paper reconceptualises the EKB model's memory stage as a contested sustainability node. This re-specification is recursive and AI-mediated, updating the model that has anchored consumer behaviour research for over five decades. The dual-pathway framework extends beyond the AI-sustainability domain: any platform environment in which AI manages consumer memory traces can be examined using this dual pathway to assess its sustainability architecture.

The paper expands the AI-AM construct in marketing research beyond cognitive efficiency to encompass its SDG implications (Grewal et al. 2025; Bergianti 2025). The paper introduces AI literacy as a key competence for sustainable consumption research. If this competence is missing, it creates a type of contextual vulnerability in AI-mediated markets (Baker et al. 2005). Therefore, SDG12 education programmes in digitally advanced markets should include AI-AM awareness as a core competency (United Nations 2015; Ng et al. 2021).

6.2 | Managerial Implications

For platform designers, the sustainability-aligned AI-AM typology (Table 3) provides eight actionable design principles. The most commercially viable starting points are need-confirmation architecture and accuracy-first review curation, as consumers

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who buy the right product the first time are less likely to return it, reducing customer service costs and waste (Das and Kunja 2024; von Zahn et al. 2025). AI transparency features (Principle 7) similarly protect consumers and enhance platform reputation; transparency about algorithmic processes increases consumer trust and reduces perceived AI manipulation (Puntoni et al. 2021; Longoni and Cian 2022). For retailers and brands, the algorithmic lock-in finding (U4) identifies a problem with direct commercial consequences: when recommendations persistently surface products the consumer has already rejected, brand equity and trust deteriorate alongside sustainability outcomes (Jain et al. 2024). For marketing educators, the finding that AI literacy is a sustainability competence supports integrating AI-AM awareness into consumer behaviour curricula (Ng et al. 2021), equipping students to make deliberate rather than algorithmically guided consumption decisions.

6.3 | Policy and Societal Implications

For platform regulators in Singapore, the findings support and extend the Personal Data Protection Commission's 2024 AI advisory guidelines on AI recommendation and decision systems (PDPC 2024). The invisible influence finding (A3) qualitatively supports the PDPC's focus on notification, accountability and explainability, showing that consumers who cannot recognise AI's impact on their decisions cannot exercise the autonomy these guidelines aim to protect. For policymakers across Asia-Pacific, the findings suggest that sustainability-focused AI regulation should be adapted to cultural contexts. Regulatory frameworks from Europe, including the EU AI Act's rules on manipulative AI and transparency (European Parliament and Council of the European Union 2024), offer a reference point, but they apply only when certain statutory thresholds are met, not to AI-mediated retail cueing in general. Adapting them for Asia-Pacific contexts would need to consider this difference.

At the societal level, the age-associated patterns observed in this sample (C3) suggest that AI-AM's consequences for sustainability may vary across consumer groups. Consumers who are less able to recognise and critically evaluate algorithmic influence may be more vulnerable to the unsustainable pathway, lacking the ability to resist it. Because invisible influence (A3) is the most concerning autonomy stance for sustainability identified in this study, public initiatives that make AI-AM's reactivation of behavioural traces visible to regular consumers (rather than only through specialist education programmes), could complement regulation. These context-based findings come from a single-market qualitative study and require testing across consumer groups and markets before wider conclusions about societal claims can be drawn.

6.4 | Limitations

Five methodological limitations should be noted, each connected to its implications for interpretation and directions for future research. First, all 30 participants are Singapore-based, so theoretical rather than statistical generalisation is claimed; cross-cultural replication in other Asia-Pacific markets is needed before the dual-pathway model can be considered regionally generalisable. Second, the protocol was not designed specifically to probe sustainability mechanisms; therefore, social desirability bias, although mitigated by the protocol design, cannot be fully eliminated. A purpose-built quantitative tool would measure these constructs directly, rather than inferring them from interviews designed for another purpose. Third, the data were collected in April–May 2025, so specific platform features and AI capabilities may have evolved since then; longitudinal tracking would reveal whether the pathways identified remain stable or shift over time. Fourth, the practitioner account in Section 4.6 is the researcher's own reflexive record, not independently audited and is offered as interpretive context alongside the consumer interviews rather than as independently verified evidence. Triangulating this account against other sellers' operational data would strengthen confidence in the described seller-side architecture. Finally, the boundary between S1 (need-confirmation cueing) and U1 (AI-triggered impulse purchase) depends on whether prior intent existed before an AI cue appeared; an independent review confirmed this judgement is conceptually sound but can be difficult to apply consistently when AI recommendations come from the participant's prior browsing history. Platform-based experiments manipulating AI-AM design features under controlled conditions would allow direct testing of this boundary instead of relying on self-reports.

6.5 | Future Research Agenda

Six priority directions follow from the findings:

1. Quantitative validation through survey-based scale development and cross-cultural SEM testing
2. Platform-based experiments manipulating AI-AM design features to test the effects of pathway shifts
3. Experimental investigation of how AI-curated social proof (C2) interacts with promotional escalation cues and unverified sustainability claims to shape consumption norms in collectivist markets.
4. Longitudinal tracking of autonomy stance development under AI-AM exposure.
5. Comparative research across multi-sellers and multi-platforms to triangulate the practitioner account and establish an empirical basis for platform-differentiated governance
6. Testing the scope of the dual-pathway model across diverse Asia-Pacific markets (e.g., Japan, Thailand, Australia).

6.6 | Conclusion

AI-augmented memory is more than just a cognitive convenience tool. The findings suggest that it influences whether consumers buy the right amount thoughtfully or buy more quickly with less thought and more waste. The dual-pathway model developed here is based on 30 semi-structured interviews with Singapore-based e-commerce consumers. Mapped

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to the EKB decision journey, it explains how the same AI mechanism can either advance or undermine SDG12. The outcome depends on AI design, consumer autonomy stance and cultural context.

The paper's key empirical finding is that invisible influence, the unseen impact of AI-AM on consumer decisions, is the most harmful autonomy stance for sustainability, because it prevents consumers from exercising the deliberative resistance that activates the sustainable pathway. Therefore, making AI's influence visible is not just about transparency; it is essential for promoting sustainability. Consumers who see the architecture can negotiate with it. Consumers who cannot comply are governed by it.

There is no easy answer to the problem this paper addresses. In many digital-retail settings, commercial incentives often privilege conversion over conservation. However, the findings show that the same AI-AM capabilities that drive overconsumption can also support responsible purchasing if configured differently. The sustainability-aligned AI-AM typology is a blueprint for that shift. Whether it is acted on depends on platform designers, regulators and consumers. As one participant said, the choice comes down to 'willpower versus AI seduction'.

Author Contributions

Jack Ng-Soon-Chye: conceptualisation, theoretical framework development, empirical data collection (30 semi-structured consumer interviews), formal analysis and thematic coding, writing – original draft, writing – review and editing, methodology and project administration. As the sole author, Jack Ng-Soon-Chye bears full responsibility for all aspects of the research and its reporting.

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Grammarly and QuillBot were used in the preparation of this manuscript for grammatical language detection and correction and for paraphrasing and sentence-level clarity improvement respectively. Both tools were used at the level of language and expression. All theoretical constructs, conceptual contributions, empirical analysis and interpretive conclusions are the intellectual work of the author. The author takes full responsibility for the integrity of the research and the accuracy of all claims.

All empirical data, comprising 30 semi-structured interviews conducted under the University of Canberra Human Research Ethics Committee approval (Reference Number: 13391), were collected, transcribed and thematically analysed exclusively by the author. No artificial intelligence tools were involved in data collection, data creation or coding decisions.

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Ethics Statement

Ethics approval was granted by the University of Canberra Human Research Ethics Committee (Number: 13391).

Consent

Written informed consent was obtained from all participants prior to each interview.

Conflicts of Interest

The author declares no conflicts of interest.

Data Availability Statement

The anonymised interview data are not publicly available due to ethics and participant-confidentiality restrictions. Selected de-identified excerpts supporting the analysis are included in the manuscript and supplementary appendices. Further access may be considered subject to ethics approval and anonymisation requirements.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Supporting Information Appendix A:** Coding framework and conceptual-sufficiency summary. **Supporting Information Appendix B:** Negative-case log. **Supporting Information Appendix C:** Independent challenge review summary.

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